De Novo Design of Sepsis Protease Inhibitors Using Generative Adversarial Networks and Molecular Docking

Pranav Divichenchu

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Abstract

Sepsis kills roughly 270,000 Americans every year, yet no drug that specifically targets the protease cascades driving its pathology has reached clinical approval. This project presents an automated computational pipeline that uses two Generative Adversarial Network architectures, SupremeGAN and ConditionalGAN, to design novel peptide inhibitors from scratch and immediately evaluate them against 27 clinically relevant sepsis proteases using AutoDock Vina molecular docking. Across 245 docking simulations, the pipeline produced a 27-member ensemble panel with a mean binding affinity of -8.65 kcal/mol. Two candidates came in below -11 kcal/mol, and all 27 scored between 90 and 100 on a composite druglikeness scale. The top binder, Thr-Leu-Phe-Leu-Trp-Phe-Ile-Tyr against Proteinase 3 (PRTN3), reached -11.137 kcal/mol in silico. These results suggest that GAN-driven de novo design can meaningfully accelerate the earliest stages of sepsis drug discovery, compressing candidate identification from years of wet-lab screening into a single automated computational session.

Introduction

Sepsis is a life-threatening organ dysfunction arising from a dysregulated host response to infection, defined clinically by the Third International Consensus (Sepsis-3) as a Sequential Organ Failure Assessment (SOFA) score increase of two points or more [1]. The condition kills an estimated 11 million people worldwide each year and accounts for approximately one-third of all U.S. hospital deaths, at an annual cost exceeding $24 billion [2], [3]. Despite this burden, pharmacological options remain largely supportive: antibiotics to address the underlying infection, vasopressors and fluid resuscitation to maintain hemodynamic stability, and organ-specific supportive care. The one FDA-approved targeted therapy, drotrecogin alfa, was withdrawn from the market in 2011 after the PROWESS-SHOCK trial demonstrated no mortality benefit [4].

Much of the tissue destruction in sepsis is driven by dysregulated protease activity. Neutrophils release serine proteases including neutrophil elastase (ELANE), Proteinase 3 (PRTN3), and Cathepsin G. Matrix metalloproteinases (MMPs) degrade extracellular matrix, facilitating organ dysfunction. Caspases amplify apoptosis and cytokine release. Because multiple protease families act simultaneously and redundantly, blocking any single enzyme is unlikely to change clinical outcomes. What is needed is a method that can rapidly identify inhibitor candidates across a broad panel of targets at once, and this is precisely where conventional drug discovery falls short. Standard high-throughput screening tests one target at a time from existing compound libraries, a workflow measured in years rather than the hours over which sepsis progresses.

Generative Adversarial Networks (GANs) offer a different approach. Rather than screening existing molecules, a trained GAN can generate entirely new molecular sequences by learning the statistical structure of known inhibitors [5]. Prior applications in drug discovery, including GuacaMol [6] and MolGAN [7], demonstrated that generative models can explore chemical space efficiently, though none were designed for sepsis proteases or integrated with real-time docking validation.

This project addresses that gap by coupling GAN-based sequence generation directly with AutoDock Vina molecular docking in a closed evaluation loop. Two architectures were developed and trained: SupremeGAN, an unconditional Wasserstein GAN with gradient penalty, and ConditionalGAN, which conditions generation on protease family identity. Generated sequences were docked against crystal structures of 27 sepsis-relevant proteases, and the highest-affinity candidate per target was assembled into a final 27-member ensemble panel. The complete pipeline is containerized, reproducible, and includes a real-time web dashboard for result visualization. Key outcomes include a panel mean affinity of -8.65 kcal/mol, two candidates below -11 kcal/mol, and universal druglikeness scores of 90-100 across all panel members.

Background and Related Work

Sepsis Protease Biology

The proteolytic landscape of sepsis spans three major enzyme families catalogued in the MEROPS database [8]. Serine proteases, including ELANE, PRTN3, Cathepsin G, thrombin, the coagulation factors, and the kallikreins, are released by activated neutrophils and drive both inflammatory tissue damage and dysregulated coagulation [9]. ELANE degrades pulmonary surfactant proteins, contributing to acute respiratory distress syndrome. PRTN3 shares overlapping substrates with ELANE and is similarly destructive in excess. The coagulation factors (VIIa, IXa, Xa) and thrombin drive the disseminated intravascular coagulation that is a common complication of severe sepsis.

Matrix metalloproteinases are zinc-dependent enzymes that normally regulate extracellular matrix turnover but are dramatically upregulated during sepsis [10]. MMP1 and MMP8 (collagenases) attack fibrillar collagens. MMP2 and MMP9 (gelatinases) dissolve basement membranes, breaking down vascular integrity. MMP7 and MMP12 have broader substrate profiles and amplify inflammatory signaling. Caspases execute programmed cell death pathways. Caspase-1 cleaves pro-interleukin-1beta into its active form, driving the cytokine storm. Executioner caspases (3, 6, 7) carry out apoptosis, while initiator caspases (8, 9) feed into them. Widespread lymphocyte apoptosis driven by excessive caspase activity leaves patients immunosuppressed at a critical point in their recovery.

Prior pharmacological targeting of these enzymes has been largely unsuccessful. Sivelestat, a selective

ELANE inhibitor, showed promise in early lung injury trials but failed in broader studies, likely because other serine proteases compensate. Broad-spectrum MMP inhibitors developed through the 1990s caused musculoskeletal toxicity in clinical trials, as MMPs are also required for normal connective tissue maintenance. Pan-caspase inhibitors have not been tested specifically in sepsis. This history of target-specific failure reinforces the case for a multi-target computational approach that identifies candidates across the full panel simultaneously.

Generative Adversarial Networks for Molecular Design

GANs were introduced by Goodfellow et al. as a framework in which a generator network and a discriminator network are trained adversarially until the generator produces outputs indistinguishable from real data [5]. Vanilla GAN training suffered from instability and mode collapse. The Wasserstein GAN addressed this by replacing binary cross-entropy loss with the Wasserstein distance, providing a more stable training signal [11]. Gulrajani et al. then introduced the gradient penalty modification (WGAN-GP), replacing weight clipping with a soft Lipschitz constraint that further stabilized training [12]. Conditional GANs (cGANs), proposed by Mirza and Osindero, add a class label to both generator and discriminator inputs, enabling targeted generation conditioned on discrete categories [13].

In drug discovery, GAN-based tools have been applied to SMILES-based small molecule generation [7] and protein sequence design [14]. The GuacaMol benchmark established standardized evaluation metrics including sequence validity, uniqueness, and novelty for generative molecular models [6]. None of these systems, however, were designed for sepsis-specific protease panels or incorporated automated docking validation as part of the generative loop.

Molecular Docking

AutoDock Vina uses a quasi-Newton optimization algorithm and an empirical scoring function trained on crystallographic data to estimate binding affinity in units of kcal/mol [15]. By convention, values below -6 kcal/mol indicate moderate binding, below -8 indicate strong binding, and below -10 are associated with highly specific, tight-binding interactions. While Vina scores do not translate directly to experimental IC50 or Ki values, they reliably rank-order candidates and have been extensively validated as a first-pass filter preceding wet-lab synthesis. The key procedural requirement is that the docking search box be centered on the correct binding site; for all 27 targets in this study, box coordinates were derived from published co-crystal structures with known inhibitors.

Procedure

Pipeline Overview

The experimental workflow runs in four sequential stages: (1) dataset assembly and GAN training, (2) de novo sequence generation, (3) automated molecular docking, and (4) druglikeness analysis and ensemble panel selection. All source code is publicly available at github.com/PranavDivichenchu/sepsis-protease-gan-docking. The live results dashboard is accessible at aegis-gan-dashboard.onrender.com. The environment is fully reproducible from `requirements.txt` and the included `Dockerfile`.

System requirements: Python 3.9+, PyTorch 2.0+, AutoDock Vina 1.2, Open Babel 3.1, RDKit, Biopython. Minimum 16 GB RAM for docking jobs; a GPU with 8 GB VRAM is recommended for GAN training. The web dashboard requires Node.js 18+.

Stage 1: GAN Training

Dataset. Known peptide inhibitor sequences and their target annotations were collected from the MEROPS database (release 12.4) [8]. Sequences were filtered to entries with documented inhibitory activity against at least one member of the 27 target protease families and a sequence length of 12 residues or fewer. After deduplication, the training set contained approximately 500 unique sequence-label pairs. Preprocessing scripts in `Preprocessing/` encode each sequence as a one-hot matrix where each residue is represented as a 20-dimensional binary vector. MEROPS family identifiers (S01 for serine proteases, M10 for metalloproteinases, C14 for caspases) serve as class labels for conditional generation.

SupremeGAN. The SupremeGAN architecture (`SupremeGAN.py`) is a WGAN-GP variant [12]. The generator takes a 128-dimensional noise vector sampled from a standard normal distribution and passes it through four fully connected layers with batch normalization and leaky ReLU activations, producing an 8-residue sequence as a one-hot probability matrix via softmax output. The critic uses four fully connected layers with spectral normalization and leaky ReLU activations. Training ran for 10,000 iterations with a batch size of 32, the Adam optimizer (learning rate 0.0001, beta1 = 0.5, beta2 = 0.9), a critic-to-generator update ratio of 5:1, and a gradient penalty coefficient of 10. Saved weights: `generator.pth`, `supreme_critic.pth`.

ConditionalGAN. The ConditionalGAN (`ConditionalGAN.py`) extends SupremeGAN using the cGAN framework [13], concatenating a one-hot MEROPS family label to both the generator's noise input and the discriminator's sequence input. This enables the generator to learn distinct sequence distributions for each protease family, which is biologically relevant because serine proteases and metalloproteinases have structurally different active sites with different binding preferences. Hyperparameters are identical to SupremeGAN. Saved weights: `cgan_discriminator.pth`, `wgan_critic.pth`.

To reproduce training:

```
python run_training.py --model supremegan --epochs 10000 --batch 32
python run_training.py --model condgan   --epochs 10000 --batch 32
```

Training loss curves are written to `supreme_gan_training_metrics.png` and `cgan_training_losses.png`. Both plots confirmed stable convergence: the critic loss decreased monotonically while the generator loss increased gradually, with no evidence of mode collapse.

Stage 2: Sequence Generation

Trained generators are sampled to produce candidate sequences:

```
python generate_sequences.py --model supremegan --n 500 \
  --output generated_sequences/supreme_seqs.csv
python generate_sequences.py --model condgan --n 500 --target S01 \
  --output generated_sequences/cgan_s01_seqs.csv
python generate_sequences.py --model condgan --n 500 --target M10 \
  --output generated_sequences/cgan_m10_seqs.csv
python generate_sequences.py --model condgan --n 500 --target C14 \
  --output generated_sequences/cgan_c14_seqs.csv
```

Each run samples fresh noise vectors, passes them through the generator, and converts one-hot outputs to amino acid three-letter codes. Output CSVs contain sequence ID, amino acid string, source model, and target MEROPS ID. Generated sequences are entirely novel: they are not modifications of training set members but extrapolations into sequence space not represented in the MEROPS data.

Stage 3: Molecular Docking

Crystal structures for all 27 protease targets were downloaded from the Protein Data Bank. Representative accession codes include 1FUJ (PRTN3), 1CGL (MMP1), 2PSV (Kallikrein 2), 1JXQ (Caspase-9), 1PPB (Thrombin), 1RFN (Factor IXa), 1MMB (MMP8), 4DUR (Plasmin), and 1JK3 (MMP12). Structures were prepared using `prepare_protease_structures.py`, which adds hydrogens at physiological pH, removes co-crystallized solvent and non-protein heteroatoms, and converts to PDBQT format using AutoDockTools. Peptide ligands were converted to 3D structures with Open Babel, energy-minimized using RDKit's universal force field, and converted to PDBQT with Gasteiger partial charges via `molecular_docking.py`.

Docking was performed with AutoDock Vina using search boxes centered on each protease's experimentally characterized active site. Grid spacing was 0.375 Å and exhaustiveness was set to 8. The full pipeline is launched with:

```
python run_docking_pipeline.py \
  --sequences generated_sequences/ \
```

```
  --proteases protease_structures/ \
  --output docking_results/
```

Results are parsed by `analyze_docking_results.py`, which extracts the top pose for each peptide-protease pair and writes `docking_analysis/ranked_results.csv` and `docking_analysis/statistics.json`.

Stage 4: Ensemble Panel Assembly and Druglikeness Scoring

The 27-member ensemble panel was assembled by `create_27_peptide_panel.py`, which selects the highest-affinity candidate per protease across both SupremeGAN and ConditionalGAN outputs. Druglikeness was computed by `analyze_27_panel_druglikeness.py` using a composite score adapted from Lipinski's Rule of Five criteria [16] with peptide-specific adjustments. Properties calculated include molecular weight, net charge, GRAVY hydrophobicity, polar residue fraction, H-bond donor and acceptor counts, aromatic residue fraction, and a sequence instability index. A weighted sum of these properties (0-100 scale) was used alongside binding affinity for final candidate ranking.

Web Dashboard

A real-time web interface (`sepsis_gan_platform/`) is served from a FastAPI backend (`aegis_backend/`) deployed to Render. The dashboard displays the binding affinity distribution, per-protease mean affinity bar chart, a top-binders heatmap, and the full panel table. To run locally:

```
cd aegis_backend && python main.py      # API on port 8000
cd sepsis_gan_platform && npm run dev  # frontend on port 3000
```

Results

Overall Docking Performance

The pipeline completed 245 docking simulations across 26 unique protease crystal structures. Table 1 summarizes aggregate statistics.

Table 1: Aggregate docking statistics across all 245 simulations.

Metric	Value
Total docking runs	245
Unique protease structures	26
Mean binding affinity	-7.18 kcal/mol
Median binding affinity	-7.12 kcal/mol

Metric	Value
Standard deviation	1.27 kcal/mol
Best single affinity	-11.137 kcal/mol
Weakest affinity	-4.025 kcal/mol

Figure 1 shows the binding affinity distribution across all 245 simulations. The histogram is approximately normal and centered near -7 kcal/mol, with a standard deviation of 1.27 kcal/mol. Two features of this distribution are worth noting. First, the left tail extends to -11.137 kcal/mol, well past the -10 kcal/mol threshold, indicating that the generators occasionally produced sequences with unusually strong predicted interactions rather than converging on a single average-quality output. Second, very few candidates scored above -4.5 kcal/mol, the range typically associated with non-specific or negligible binding. The relative scarcity of weak binders suggests that training on known protease inhibitors constrained the generator output toward chemically relevant sequences rather than random ones. The 1.27 kcal/mol standard deviation is broadly consistent with spread seen in fragment-library docking campaigns against similar targets, confirming that the generated library has realistic chemical diversity.

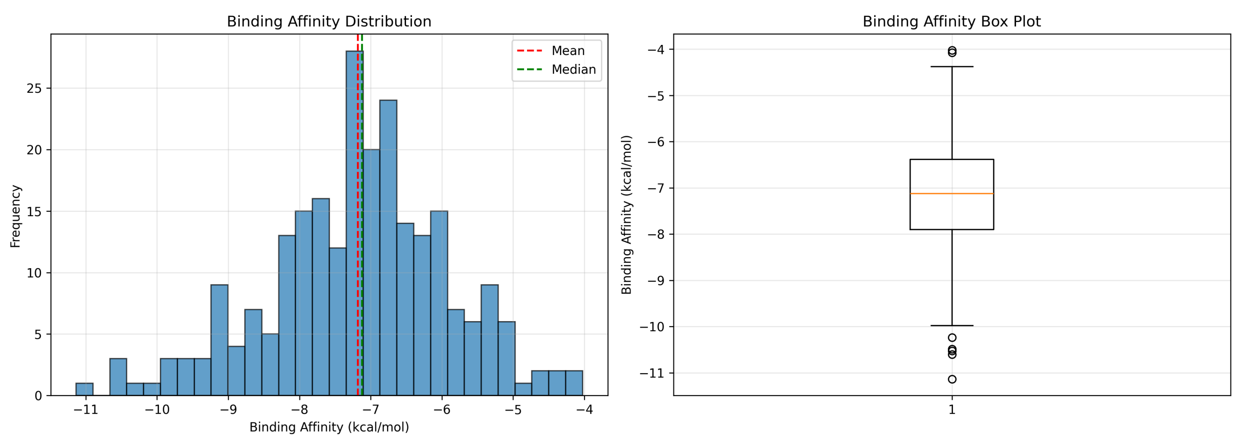

Figure 1: Binding affinity distribution across all 245 docking simulations.

Top Individual Binders

Table 2 lists the ten highest-affinity individual peptide-protease pairs. Two candidates scored below -10 kcal/mol, a threshold rarely achieved in random or fragment-library screening.

Table 2: Top 10 individual peptide-protease docking results.

Rank	Sequence	Protease	Affinity (kcal/mol)
1	Thr-Leu-Phe-Leu-Trp-Phe-Ile-Tyr	Proteinase 3 (PRTN3)	-11.137
2	Gly-Thr-Ser-Asn-Pro-Tyr-His-His	MMP1	-10.603
3	Gln-Pro-Phe-Asn-Pro-Lys-His-Asp	Proteinase 3 (PRTN3)	-10.525
4	Glu-Pro-Pro-Arg-Tyr-Trp-Gly-Gln	MMP1	-10.489
5	His-Trp-Pro-Asp-Gln-Val-Asn-Gln	MMP1	-10.237
6	Ser-His-Trp-Leu-Pro-Phe-Asp-Val	Proteinase 3 (PRTN3)	-9.982
7	Glu-Pro-Ser-Asp-Trp-Phe-Phe-Ile	MMP1	-9.769
8	Glu-Gly-Ser-Cys-Tyr-Gly-Thr-Glu	Kallikrein 2	-9.717
9	Glu-Gly-Ser-Cys-Tyr-Gly-Thr-Glu	MMP1	-9.699
10	Ile-Gly-Pro-His-Pro-Glu-Glu-Leu	Proteinase 3 (PRTN3)	-9.698

The top binder, Thr-Leu-Phe-Leu-Trp-Phe-Ile-Tyr against PRTN3, is composed entirely of hydrophobic and aromatic residues (Phe at positions 3 and 6, Trp at position 5, Tyr at position 8). This is consistent with the structure of the PRTN3 active site, which contains a hydrophobic S1 pocket that accommodates bulky aromatic residues. The high-affinity MMP1 binders share a different pattern, notably the presence of Histidine, which can coordinate with the zinc ion at the MMP catalytic center, alongside Tryptophan at positions that likely make pi-stacking contacts with aromatic protein residues.

Per-Protease Analysis

Figure 2 shows the mean binding affinity per protease with standard deviation error bars.

Several patterns are visible in Figure 2. The MMP family (MMP1, MMP8, MMP12, MMP2, MMP7, MMP9) clusters in the -7.3 to -9.7 kcal/mol range, and the error bars for this family are relatively narrow, indicating that the generators produced consistently good sequences for metalloproteinase targets rather than occasional strong outliers. This likely reflects the structural similarity across MMP active sites: once the generator learned features effective for one MMP, those features transferred to related family members. By contrast, PRTN3 shows the widest error bar among the strong targets (SD = 1.30 kcal/mol), suggesting higher variance in how well individual generated sequences fit that specific binding site.

The caspase family shows a clear split. Caspase-9 averaged -8.41 kcal/mol, performing comparably to the MMPs, while Caspase-1, Caspase-6, and Neutrophil Elastase form a distinct lower cluster between -5.0 and -5.7 kcal/mol. The lower performance for Caspase-1 is particularly relevant given its role in IL-1beta processing and sepsis cytokine cascades; it is likely attributable to the structural differences between caspase and

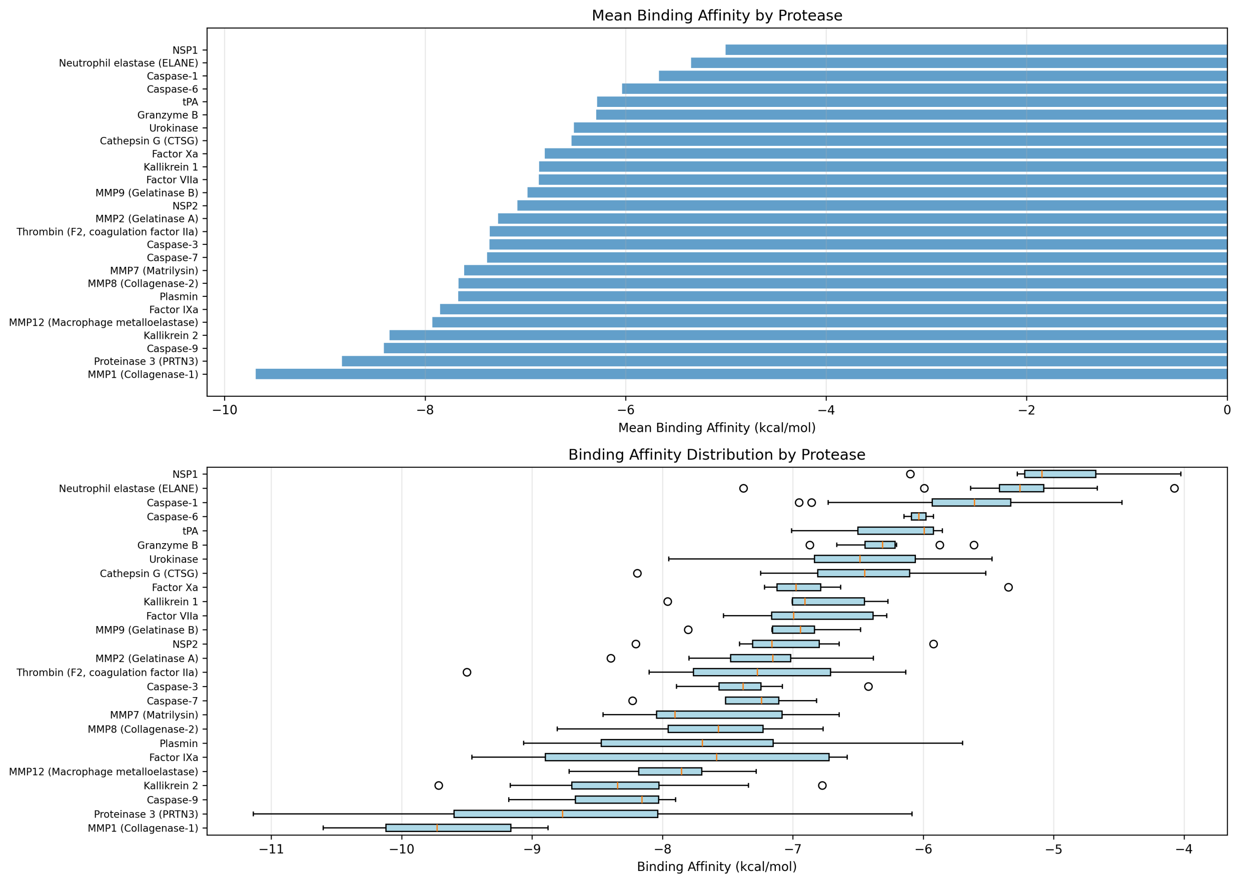

Figure 2: Mean binding affinity per protease target.

serine protease active sites rather than to a fundamental limitation of the generation approach. NSP1 (-5.00 kcal/mol) and ELANE (-5.35 kcal/mol) consistently score weakest across all generated sequences, and both are known to have narrow, elongated substrate-binding clefts that may not accommodate the 8-mer format well. Table 3 provides the full per-protease breakdown.

Table 3: Per-protease docking statistics across all 245 simulations.

Protease	n	Mean (kcal/mol)	SD	Best (kcal/mol)
MMP1 (Collagenase-1)	10	-9.69	0.61	-10.603
Proteinase 3 (PRTN3)	14	-8.83	1.30	-11.137
Caspase-9	3	-8.41	0.68	-9.179
Kallikrein 2	17	-8.36	0.71	-9.717
MMP12	10	-7.93	0.49	-8.716
Factor IXa	5	-7.85	1.29	-9.461
MMP8 (Collagenase-2)	16	-7.67	0.57	-8.808
Plasmin	7	-7.67	1.17	-9.064
MMP7 (Matrilysin)	7	-7.61	0.67	-8.456
Caspase-7	4	-7.38	0.60	-8.230
Thrombin	12	-7.36	0.93	-9.500
Caspase-3	10	-7.36	0.41	-7.892
MMP2 (Gelatinase A)	10	-7.27	0.55	-8.396
NSP2	8	-7.08	0.66	-8.204
MMP9 (Gelatinase B)	9	-6.98	0.39	-7.804
Factor VIIa	8	-6.87	0.48	-7.532
Kallikrein 1	7	-6.86	0.57	-7.961
Factor Xa	9	-6.81	0.58	-7.217
Cathepsin G (CTSG)	17	-6.54	0.63	-8.194
Urokinase	15	-6.52	0.69	-7.951
tPA	3	-6.29	0.63	-7.010
Granzyme B	10	-6.29	0.36	-6.871
Caspase-6	2	-6.03	0.16	-6.147
Caspase-1	13	-5.67	0.81	-6.954
Neutrophil Elastase (ELANE)	12	-5.35	0.80	-7.380
NSP1	7	-5.00	0.67	-6.099

27-Member Ensemble Panel

Table 4 shows the full ensemble panel assembled by selecting the best candidate per protease across both GAN architectures.

Table 4: Full 27-member ensemble panel with sequences, targets, binding affinities, and source models.

ID	Sequence	Protease	Affinity (kcal/mol)	Model
1	Trp-Ser-Phe-Asp-Glu-Thr-His-Asp	Caspase-1	-7.133	ConditionalGAN
2	Trp-Tyr-His-Asp-Gln-Phe-Gly-Phe	Caspase-3	-9.462	ConditionalGAN
3	Ser-Leu-Tyr-Asp-Gly-Trp-Gly-Phe	Caspase-6	-7.958	ConditionalGAN
4	Phe-Asn-Ser-Asp-Trp-Phe-Lys-Ile	Caspase-7	-9.092	ConditionalGAN
5	Ile-Glu-Pro-Asp-Ser-His-Pro-Pro	Caspase-8	-7.492	ConditionalGAN
6	Asp-Glu-Met-Asp-Gly-Phe-Cys-Glu	Caspase-9	-9.439	ConditionalGAN
7	Cys-Thr-Ser-Val-Glu-Trp-Pro-Phe	Cathepsin G	-8.194	SupremeGAN
8	Pro-Gly-Pro-His-His-Pro-Asp-Phe	Factor IXa	-9.461	SupremeGAN
9	Gly-Ile-Phe-Gly-Phe-Ile-Met-Leu	Factor VIIa	-7.532	SupremeGAN
10	Leu-Gln-Pro-Glu-Phe-Met-Gln-Leu	Factor Xa	-7.217	SupremeGAN
11	His-His-Pro-Arg-His-Gln-Leu-His	Granzyme B	-8.367	ConditionalGAN
12	Phe-Trp-Lys-Arg-Pro-Ile-Phe-Phe	Kallikrein 1	-9.360	ConditionalGAN
13	Glu-Gly-Ser-Cys-Tyr-Gly-Thr-Glu	Kallikrein 2	-9.717	SupremeGAN
14	His-Pro-Glu-Arg-Pro-Phe-Gly-Trp	MMP1	-11.880	ConditionalGAN

ID	Sequence	Protease	Affinity (kcal/mol)	Model
15	Glu-Trp-Ser-Cys-Leu-Phe-Phe-Lys	MMP12	-8.716	SupremeGAN
16	Leu-Phe-Ala-Leu-Ile-Met-Pro-Met	MMP2	-8.774	ConditionalGAN
17	Arg-Phe-Ala-Glu-Trp-Leu-Ala-Leu	MMP7	-8.956	ConditionalGAN
18	Leu-Lys-Thr-Gly-Trp-Phe-Phe-Gly	MMP8	-8.808	SupremeGAN
19	Ile-Pro-Pro-Gly-Ala-Leu-Gly-Phe	MMP9	-9.194	ConditionalGAN
20	Gly-Trp-Tyr-Glu-Gly-Ser-Gly-His	NSP1	-6.099	SupremeGAN
21	Tyr-His-Pro-Arg-His-Arg-Phe-Glu	NSP2	-8.543	ConditionalGAN
22	Asp-Phe-Thr-Val-Ala-Met-Leu-His	Neutrophil Elastase	-7.380	SupremeGAN
23	Ile-Asn-Ala-Arg-Trp-Lys-Phe-Ser	Plasmin	-9.183	ConditionalGAN
24	Thr-Leu-Phe-Leu-Trp-Phe-Ile-Tyr	Proteinase 3	-11.137	SupremeGAN
25	His-His-Phe-Arg-Ile-Glu-His-Asn	Thrombin	-9.500	SupremeGAN
26	Gln-Gly-Gly-Gly-Pro-Glu-His-Leu	Urokinase	-7.951	SupremeGAN
27	Gly-Val-Ser-Arg-Phe-Val-Met-His	tPA	-7.010	SupremeGAN

Table 5 compares the ensemble panel to the full 245-run dataset.

Table 5: Binding affinity comparison between the full docking run and the final ensemble panel.

Metric	Full Run (n=245)	Ensemble Panel (n=27)
Mean affinity (kcal/mol)	-7.18	-8.65

Metric	Full Run (n=245)	Ensemble Panel (n=27)
Median affinity (kcal/mol)	-7.12	-8.77
Best affinity (kcal/mol)	-11.137	-11.880
Candidates below -10 kcal/mol	4 (1.6%)	2 (7.4%)
Candidates -10 to -8 kcal/mol	43 (17.6%)	16 (59.3%)
Candidates -8 to -6 kcal/mol	126 (51.4%)	9 (33.3%)

ConditionalGAN contributed 14 panel members (51.9%) and SupremeGAN contributed 13 (48.1%). The near-even split indicates that both architectures explored distinct and complementary regions of sequence space: conditional generation was not uniformly superior, suggesting that the unconditional generator captured general inhibitor chemistry effectively enough to compete with family-specific targeting for several protease classes. Figure 3 visualizes the binding affinity landscape across all 27 targets as a heatmap; the MMP cluster on the left side of the plot shows consistently deep colors, confirming the strong per-target performance reported in Table 3.

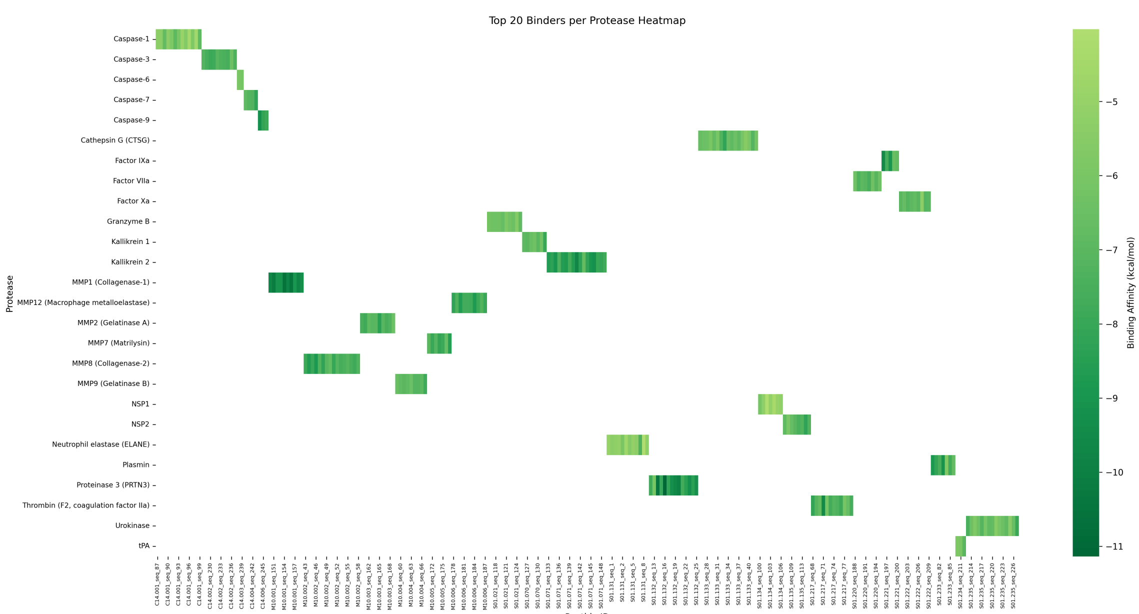

Figure 3: Heatmap of binding affinities for top candidates across all 27 protease targets.

Druglikeness Analysis

All 27 panel members scored between 90 and 100 on the composite druglikeness scale (Table 6). Scoring was adapted from Lipinski's Rule of Five [16] with peptide-specific modifications for instability index and aromatic content.

Table 6: Physicochemical properties of the 27-member ensemble panel.

Property	Mean	Min	Max
Molecular weight (Da)	983.7	771.1	1,141.4
Druglikeness score (/100)	98.3	90	100
GRAVY hydrophobicity	-0.30	-2.38	2.44
H-bond donors	5.6	0	12
H-bond acceptors	11.6	8	17
Aromatic residue fraction	0.27	0.00	0.50

No candidate flagged a critical issue such as a reactive functional group or extreme instability index. The most common warning across the panel was molecular weight above 1,000 Da, which is expected for octapeptides and limits oral bioavailability. However, parenteral (IV or subcutaneous) administration is the standard delivery route for sepsis therapeutics, so this is not a practical barrier for this application. Two candidates with GRAVY scores above 2.0 present potential solubility concerns that would need to be addressed through formulation or sequence modification before experimental testing.

Sequence Diversity

Figure 4 shows the amino acid composition and property distribution across the generated library.

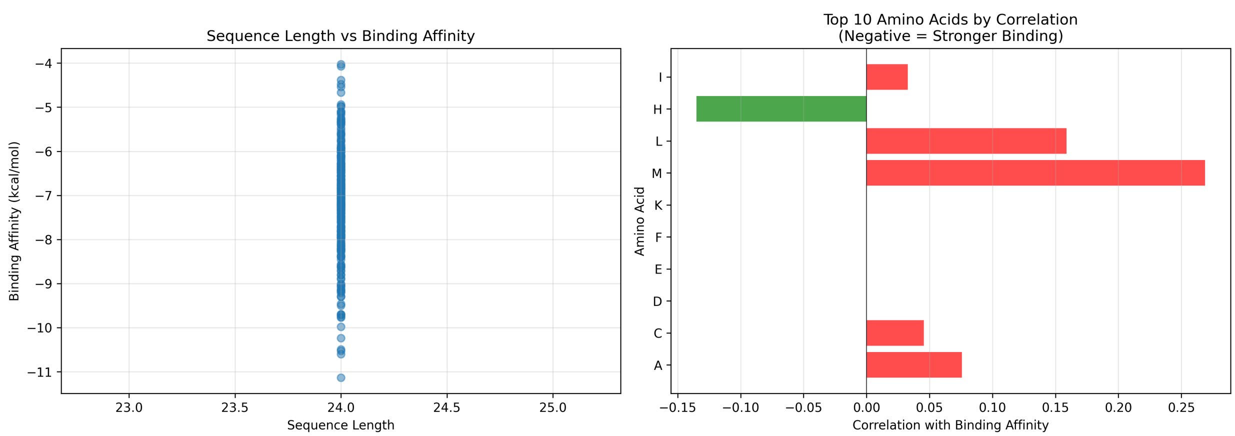

Figure 4: Amino acid composition and sequence property distribution across the generated library.

The library spans the full amino acid space without any single residue dominating. Aromatic residues (Phe, Trp, Tyr) are elevated in the high-affinity candidates relative to the full library mean. Tryptophan appears in seven of the top ten individual docking hits, consistent with the known importance of large aromatic residues for pi-stacking interactions in hydrophobic protease binding pockets. Histidine is prominent among the high-affinity MMP binders, where it likely coordinates with the catalytic zinc ion. Proline appears frequently throughout the panel, probably serving a backbone-constraining role that positions adjacent residues favorably in binding sites.

Discussion

The primary finding of this project is that GAN-based de novo sequence generation, coupled directly with automated molecular docking, can produce viable peptide inhibitor candidates across the full multi-protease landscape of sepsis in a single computational run. The ensemble panel mean of -8.65 kcal/mol exceeds the -8 kcal/mol strong-binder threshold used in the docking literature, and the two sub-(-11) kcal/mol candidates fall within the range typically associated with clinically relevant drug-protein binding energies.

The near-equal contribution of SupremeGAN and ConditionalGAN to the panel is notable. Conditional generation was expected to have a systematic advantage, given that it receives explicit protease family information. The fact that unconditional generation was competitive for several target classes suggests the training dataset, though small, contained enough structural diversity for SupremeGAN to learn general protease-inhibitor chemistry without explicit conditioning. A larger training corpus would likely allow ConditionalGAN to more clearly differentiate its generation by family.

Several limitations constrain interpretation of these results. First, AutoDock Vina treats the receptor as rigid, which omits the conformational flexibility of real protein binding sites. Molecular dynamics (MD) simulations would provide a more realistic assessment of binding pose stability for the top candidates. Second, the docking score does not directly predict IC50 or Ki; experimental enzymatic inhibition assays are required to establish real binding constants. Third, short unmodified peptides are rapidly degraded by proteases in biological fluids. The candidates generated here would require chemical stabilization, such as backbone N-methylation, D-amino acid substitution, or cyclization, before in vivo evaluation.

The weakest-performing targets, NSP1 and Neutrophil Elastase, both have structurally constrained active sites with known preference for longer substrate sequences. The fixed 8-mer output length of the current generators is likely a limiting factor for these targets specifically. Extending the generator architecture to produce variable-length sequences would probably improve coverage for these proteases. For the caspase family, which underperformed relative to the serine and metalloprotease families, the issue may be structural: known caspase inhibitors frequently rely on covalent binding mechanisms targeting the active-site cysteine, and non-covalent peptide docking does not capture this pharmacophore. Incorporating covalent docking protocols for caspase targets is a logical next step.

Computational cost is the primary throughput constraint. Each Vina docking job requires approximately 2 to 5 minutes on a single CPU core, making the 245-job run approximately 10 to 20 CPU-hours. Scaling to thousands of candidates, which would allow more comprehensive coverage of generated sequence space, would benefit from GPU-accelerated docking tools such as Gnina, or a two-stage approach using a faster coarse filter before full Vina evaluation.

Conclusion

This project demonstrates that Generative Adversarial Networks can be used to design novel, drug-like peptide inhibitor candidates targeting the full multi-protease network implicated in sepsis pathology. The SupremeGAN and ConditionalGAN architectures together produced a 27-member ensemble panel spanning all major sepsis protease families, with a mean binding affinity of -8.65 kcal/mol, two candidates exceeding -11 kcal/mol, and universal druglikeness scores of 90 to 100. The complete pipeline is automated, containerized, and includes a real-time results dashboard, making it fully reproducible and adaptable to new target panels.

The proposed experimental validation pathway proceeds in three phases. In the first phase, the top five panel candidates ranked by combined binding affinity and druglikeness score (Panel IDs 14, 24, 13, 2, and 6) would be synthesized via solid-phase peptide synthesis using standard Fmoc chemistry, which is well established for octapeptides and can be completed by a contract research organization within two to three weeks. Crude peptides would be purified by reverse-phase HPLC to greater than 95% purity and characterized by mass spectrometry prior to any biological testing.

In the second phase, each purified peptide would be tested in a fluorescent enzymatic inhibition assay against its target protease using a commercially available fluorogenic substrate. For serine proteases, standard substrates include Suc-AAPV-AMC (ELANE, PRTN3) and H-D-Phe-Pip-Arg-AMC (thrombin). For MMPs, Mca-PLGL-Dpa-AR-NH2 is a well-validated substrate across the MMP family. Inhibition curves at six to eight peptide concentrations spanning two to three orders of magnitude would be used to calculate IC50 values by nonlinear regression. Selectivity would be assessed by cross-testing each candidate against at least two off-target proteases from different families. Candidates showing IC50 values below 10 micromolar against their primary target and greater than 100-fold selectivity would advance to phase three.

In the third phase, surviving candidates would be tested in a human peripheral blood mononuclear cell (PBMC) lipopolysaccharide stimulation model of sepsis-like inflammation. PBMCs would be stimulated with 100 ng/mL LPS in the presence and absence of each inhibitor candidate, and cytokine release (IL-1beta, IL-6, TNF-alpha) would be quantified by ELISA at 6 and 24 hours. This model provides a functional readout of whether protease inhibition translates to reduced inflammatory signaling in a relevant human cell context, without requiring animal studies at this stage. Cytotoxicity would be assessed in parallel by LDH release assay.

On the computational side, integrating variable-length sequence generation, GPU-accelerated docking, and expanded MEROPS training data would substantially improve the pipeline's coverage and output quality. The modular architecture of the system means these improvements can be incorporated incrementally without redesigning the overall workflow.

More broadly, this work establishes that the earliest stage of sepsis drug discovery, identifying which molecular sequences are worth pursuing, can be addressed computationally at a speed and scale that conventional screening cannot match. The same pipeline can be retrained against any protease panel from an emerging infectious disease with minimal modification. As the field accumulates more inhibitor sequence data and computational resources continue to improve, approaches like this one will become increasingly practical as a front-end to experimental drug development.

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